

+Global Events, Commodities and Currency Markets

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Applied Data Analytics and Market Research Assignment

Based on: Brent Oil, Gold, Silver and Global Currency Tracker datasets

Assignment idea

Students will first describe what happened in the data, then connect those patterns to major global events such as the Hormuz crisis, energy supply shocks, gold accumulation, sanctions, tariffs, BRICS settlement discussions, and currency market changes.

Assignment Overview

Item	Description
Main aim	Use the provided datasets to produce simple, evidence based insights about commodities and global currency markets.
Part I	Data Analytics Component: clean, summarise, visualise and interpret the datasets.
Part II	Market Research Component: explain how global events may help interpret the observed market movements.
Expected output	A short report with charts, tables, answers to selected questions and a final business style recommendation.
Suggested length	8 to 10 pages excluding appendix.
Submission items	Report file, spreadsheet workbook, and optional 3 to 5 slide summary.

Provided Data Files

Dataset	Main fields to use	Simple questions it supports
Brent Oil.csv	Date, Value	Oil price trend, highest and lowest periods, crisis comparison
Gold 100years.csv	Date, Value	Gold long term trend, safe haven behaviour, annual change

Dataset	Main fields to use	Simple questions it supports
silver 100 years.csv	Date, Value	Silver trend, volatility compared with gold
swift_currency_tracker_all_reports.csv	Report month, data month, metric, category, currency/economy, value, unit, RMB rank, notes	Currency shares, rankings, RMB/CNY role, offshore RMB markets

Business Scenario

You are working as a junior market analyst for a financial research team. The team wants a simple but clear report showing how commodity prices and currency indicators changed over time.

Your report should not only show charts and numbers. It should also explain whether major global events may have influenced oil, gold, silver and currency market behaviour.

The final recommendation should be written for a non technical manager who wants to understand risk, opportunity and market direction.

Part I: Data Analytics Component

This part focuses on describing and visualising the data. Select the questions below and answer them with clear tables, charts and short explanations.

Minimum requirement: Answer at least 10 questions from Part I. You must include at least one question from each section: data preparation, descriptive analytics, visual analytics and comparison.

A. Data Preparation and Understanding

No.	Question	Expected output
1	What is the start date and end date available in each dataset?	Data summary table
2	How many rows are available for Brent Oil, Gold, Silver and the Swift Currency Tracker data?	Data summary table
3	Are there any missing values or repeated dates? Summarise what you found.	Data quality table
4	Convert the Date column into a proper date format	Short cleaning note

No.	Question	Expected output
	and create Month and Year columns for the commodity datasets.	
5	For the Swift dataset, identify the main metrics available, such as global payment share, trade finance share or offshore RMB economy share.	Short list or table

B. Descriptive Analytics

No.	Question	Expected output
6	What were the highest and lowest Brent Oil prices in the dataset, and in which years did they occur?	Summary table
7	What were the highest and lowest Gold prices in the dataset, and in which years did they occur?	Summary table
8	What were the highest and lowest Silver prices in the dataset, and in which years did they occur?	Summary table
9	Calculate the average price by year for Brent Oil, Gold and Silver.	Annual average table
10	Which five years had the highest average Brent Oil price?	Top five table
11	Which five years had the highest average Gold price?	Top five table
12	Which five years had the highest average Silver price?	Top five table

C. Trend and Visual Analytics

No.	Question	Expected output
13	Create a line chart for Brent Oil prices over time and describe the main periods of increase and decline.	Chart plus 3 to 5 lines of interpretation

No.	Question	Expected output
14	Create a line chart for Gold prices over time and describe the long term trend.	Chart plus 3 to 5 lines of interpretation
15	Create a line chart for Silver prices over time and describe the main peaks.	Chart plus 3 to 5 lines of interpretation
16	Create one combined chart showing Gold and Silver trends. What similarities or differences can you observe?	Combined chart and short comparison
17	Create a bar chart showing average annual Brent Oil prices for the most recent 10 years in the dataset.	Bar chart and short explanation
18	Create a simple table showing the latest available value for Brent Oil, Gold and Silver.	Latest values table

D. Currency Tracker Analytics

No.	Question	Expected output
19	Using the Swift Currency Tracker file, identify the top currencies by global payment share in the most recent available month.	Ranking table
20	What is the USD share in the most recent available global payment data?	Short answer with value
21	What is the CNY/RMB share and rank in the most recent available global payment data?	Short answer with value and rank
22	Which currencies appear most often in the top five across the available reports?	Count table
23	Which offshore RMB economy has the highest share in the available offshore RMB data?	Short answer with value
24	Create a simple bar chart for the top 10 currencies or economies from one selected month.	Bar chart and short explanation

E. Simple Comparison Questions

No.	Question	Expected output
25	Compare the latest Gold value with its average value across the full dataset. Is it above or below the long term average?	Short comparison table
26	Compare the latest Brent Oil value with its average value across the full dataset. Is it above or below the long term average?	Short comparison table
27	Which commodity appears to have the largest visible price swings: Brent Oil, Gold or Silver? Support your answer with one chart or summary table.	Chart or summary table
28	Choose one year with a strong oil price movement and check whether gold or silver also changed noticeably in that year.	Short comparison paragraph
29	Create a short dashboard page with 4 to 6 visuals summarising the most important findings.	Dashboard page
30	Write three short insights that a manager could understand without technical knowledge.	Three bullet points

Optional Bonus Analytics Questions

These questions are optional and can be used for extra marks. They are not required for passing the assignment.

Bonus no.	Question
B1	Calculate monthly or yearly percentage change for Gold, Silver and Brent Oil.
B2	Create a simple correlation table between Gold, Silver and Brent Oil using annual average prices.
B3	Create a heat map based on the correlation table.
B4	Create a scatter plot comparing Gold and Silver annual average prices.

Bonus no.	Question
B5	Use a moving average line to smooth the Gold or Brent Oil trend and explain what it shows.
B6	Create a simple forecast using a trendline and clearly state that it is only an estimate, not a guaranteed prediction.

Part II: Market Research Component

This part explains the possible connection between the observed data patterns and the global events described in the provided reports and presentations.

Minimum requirement: Select at least 4 market research questions from Part II. Your answers should use evidence from the provided files and should be linked to at least one chart or table from Part I.

A. Event Identification

No.	Question	Expected output
1	Identify three major global events mentioned in the files that could affect commodity or currency markets.	Short evidence based paragraph
2	Create a simple timeline of selected events such as the Hormuz crisis, energy supply disruption, sanctions, tariff discussion, BRICS Pay, or gold accumulation.	Timeline table
3	Which event seems most directly connected to Brent Oil prices? Explain why.	Short paragraph linked to one chart
4	Which event seems most directly connected to Gold or Silver prices? Explain why.	Short paragraph linked to one chart

B. Linking Events to Commodity Data

No.	Question	Expected output
5	How could an oil supply shock or shipping	Short explanation linked to chart

No.	Question	Expected output
	disruption affect Brent Oil prices? Link your answer to one Brent Oil chart.	
6	Why might Gold increase during geopolitical uncertainty? Link your answer to one Gold chart.	Short explanation linked to chart
7	Is Silver behaving more like a precious metal or an industrial commodity in your selected period? Explain using your chart.	Short comparison paragraph
8	Choose one crisis period and compare the behaviour of Oil, Gold and Silver in that period.	Comparison table and short explanation

C. Linking Events to Currency Data

No.	Question	Expected output
9	How does the Global Currency Tracker show the role of USD in international payments?	Short paragraph with one value or ranking
10	What does the tracker show about the role of CNY/RMB in global payments and trade finance?	Short paragraph with one value or ranking
11	Why might sanctions, alternative settlement systems, or BRICS Pay discussions matter for currency market analysis?	Short evidence based paragraph
12	What does the offshore RMB data suggest about the role of Hong Kong and other financial centers?	Short paragraph with one value or ranking

D. Business Interpretation and Recommendation

No.	Question	Expected output
13	From your analysis, which market looks most sensitive	Short argument supported by chart or table

No.	Question	Expected output
	to global events: Oil, Gold, Silver or Currencies?	
14	Write a short risk note for a company exposed to oil prices.	100 to 150 words
15	Write a short risk note for a company holding assets in USD, Gold or RMB/CNY.	100 to 150 words
16	Give three practical recommendations for investors, companies, or policymakers based on your findings.	Three recommendations

Suggested Report Structure

Section	What to include
1. Introduction	Briefly introduce the purpose of the assignment and the datasets used.
2. Data Overview	Describe the files, date ranges, variables and any simple cleaning performed.
3. Data Analytics Findings	Present the main tables and charts from Part I with clear explanations.
4. Market Research Findings	Explain the selected global events and connect them to the data patterns.
5. Final Recommendation	Provide a short business style conclusion with 3 practical recommendations.
6. Appendix	Include extra tables, screenshots, calculation notes and optional bonus analysis.

Marking Rubric

Criteria	Weight	What is assessed
Data preparation and understanding	15%	Correct use of datasets, date handling, summaries and basic data quality checks.
Charts, tables and descriptive analytics	30%	Clear visuals, correct calculations and

Criteria	Weight	What is assessed
Currency tracker analysis	15%	meaningful descriptions of trends. Correct interpretation of payment shares, currency rankings and offshore RMB evidence.
Market research connection	25%	Clear link between global events and observed market behaviour.
Final recommendation and communication	15%	Simple, professional writing with practical recommendations and well organised report structure.

Submission Checklist

- ☐ Part I questions answered.
- ☐ Part II questions answered.
- ☐ charts included.
- ☐ summary tables included.
- ☐ final dashboard or summary page included.
- ☐ global events discussed.
- ☐ Charts are labelled with clear titles and axis names.
- ☐ Final recommendation is written in plain business language.